

CLOSED-LOOP AUTO-CALIBRATION

Making the Qubit Tune Itself

Q-BLUEPRINT QUANTUM HACKATHON 2026

TEAM

SHINRAI



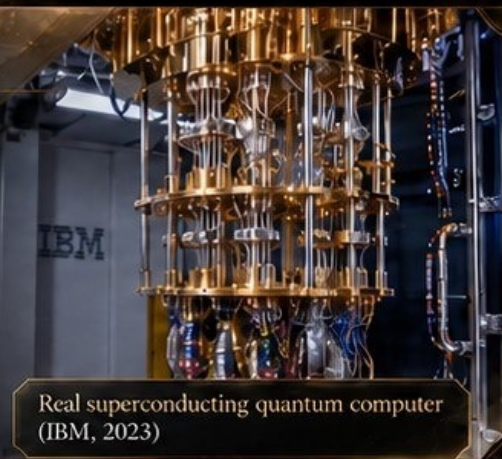
The Physical Plant

Modeling the Drifting Transmon

Our simulator models a realistic three-level transmon whose parameters drift over time due to environmental noise and hardware imperfections.

What is a Transmon?

- ◆ A type of superconducting qubit used in real quantum computers (IBM, Google, Rigetti etc.).
- ◆ Behaves like an artificial atom with quantized energy levels.
- ◆ We model the first three levels: $|0\rangle$, $|1\rangle$, $|2\rangle$.



Real superconducting quantum computer (IBM, 2023)

Three-Level Model

- ◆ We simulate a 3-level transmon in a rotating frame.
- ◆ Hamiltonian includes anharmonicity ($\alpha/2\pi = -300$ MHz) to capture leakage to level $|2\rangle$.
- ◆ This allows us to study realistic gate errors.



Energy levels of a transmon (3-level model)

Why Do Qubits Drift?

- ◆ Two main sources of drift in real devices:



1. Material Noise (TLS defects)

Microscopic defects in materials cause random fluctuations.



2. Drive Electronics Drift

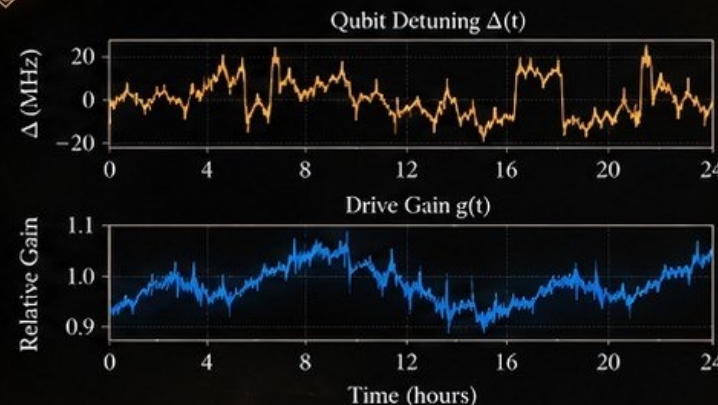
Temperature changes affect the control hardware, leading to pulse strength variations.

Drift Model Used in Simulation

- ◆ Detuning $\Delta(t)$: Modeled as an Ornstein-Uhlenbeck (OU) process + random telegraph noise (to capture sudden jumps).
- ◆ Drive gain $g(t)$: Modeled as an OU process with a sinusoidal diurnal (24-hour) variation (to mimic temperature drift).
- ◆ These models reflect realistic behaviour observed in superconducting quantum devices.



Example 24-Hour Drift Trace



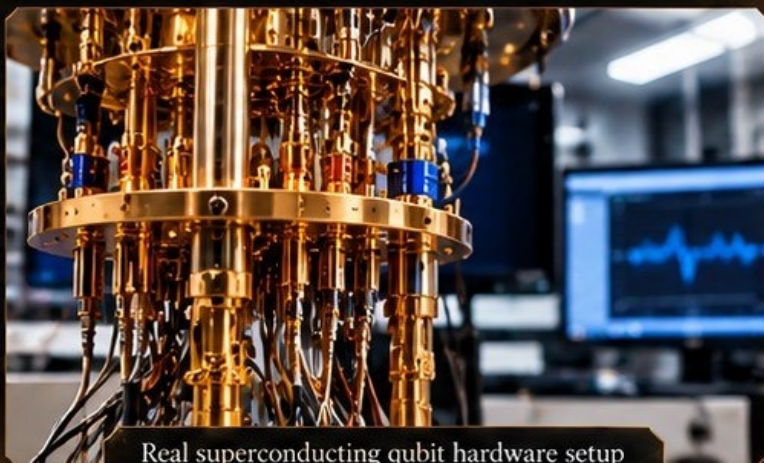
Simulated 24-hour drift of qubit detuning and drive gain

Calibration Primitives: Amplitude & DRAG

We use amplitude calibration with Rabi experiments and DRAG pulses to reduce leakage to the higher level ($|2\rangle$) in a three-level transmon.

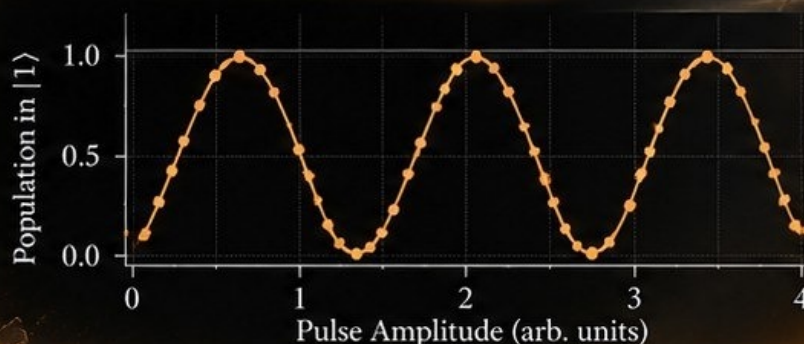
Amplitude Calibration (Rabi)

- ◆ We apply a train of π pulses and measure the qubit response (Rabi oscillations).
- ◆ This amplifies small pulse amplitude errors, making them easy to detect.
- ◆ We tune the drive gain to reach the correct π rotation (max population in $|1\rangle$).

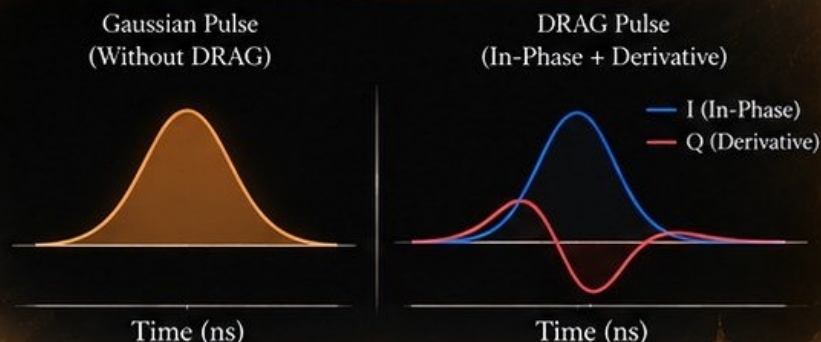


Real superconducting qubit hardware setup
(IBM, 2023)

Rabi Oscillations (Simulation)



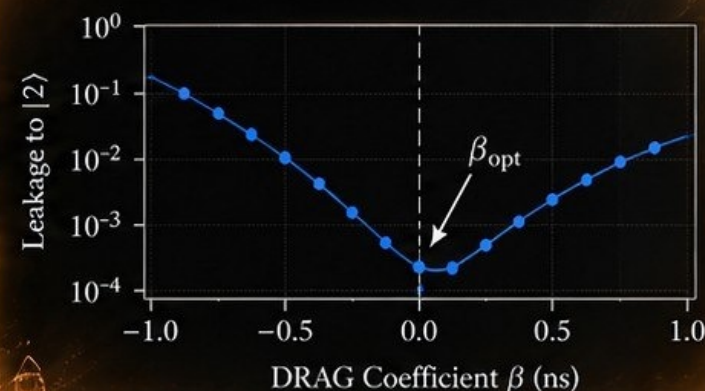
Pulse Shapes



Why DRAG?

- ◆ Fast pulses can unintentionally excite the $|1\rangle \rightarrow |2\rangle$ transition, causing leakage.
- ◆ DRAG (Derivative Removal by Adiabatic Gate) adds a derivative component to the quadrature pulse to cancel this leakage.
- ◆ We sweep the DRAG coefficient β to find the optimal value $\beta_{\text{opt}} \approx 1/|\alpha|$.

DRAG Scan (Simulation)



Ramsey Interferometry & Aliasing

We use a Ramsey sequence to estimate the qubit frequency (detuning) and a ladder of delays to resolve the aliasing problem with high precision.

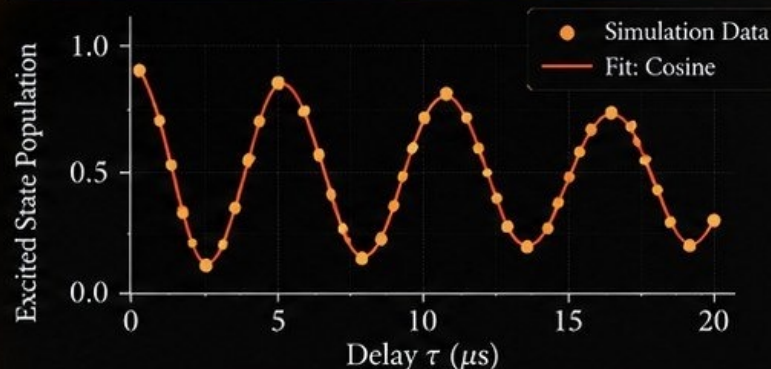
1. Ramsey Sequence

- ◆ A $\pi/2$ — wait (τ) — $\pi/2$ sequence creates an interference pattern (Ramsey fringe).
- ◆ The measured qubit population depends on the detuning Δ .
- ◆ It allows us to estimate the qubit frequency with high precision.



$$P(|1\rangle) = \frac{1}{2} [1 + C \cos(\Delta\tau + \phi)]$$

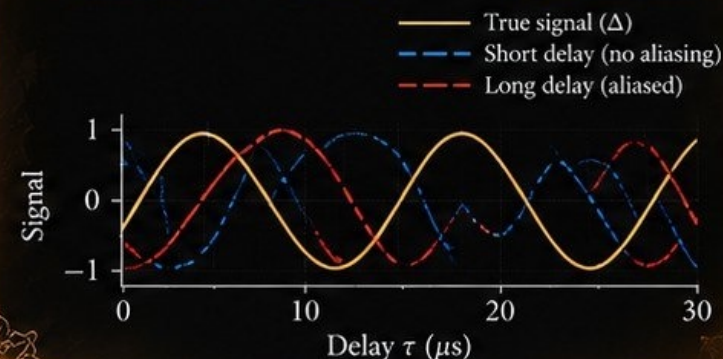
2. Ramsey Fringe (Simulation)



- ◆ The population oscillates with frequency Δ .
- ◆ From the fitted oscillation frequency, we estimate the qubit detuning.
- ◆ Longer delays give higher precision but are prone to aliasing.

3. The Aliasing Problem

- ◆ The measured phase is periodic (wraps around 2π).
- ◆ If the true detuning is large, long delays can give the wrong (aliased) frequency.
- ◆ Short delays avoid aliasing but have higher uncertainty.

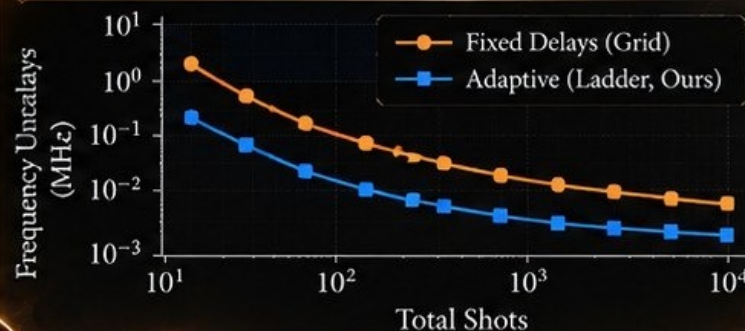


4. Ladder of Delays (Solution)

- ◆ We use a set of increasing delays: $\tau_1, \tau_2, \tau_3, \dots$
- ◆ Start with short delays to get a coarse, non-aliased estimate.
- ◆ Then use longer delays to refine the estimate and resolve the correct phase wrap.
- ◆ Combines the advantages of both short and long delays.



5. Estimation Precision



6. Final Estimated Frequency

- ◆ Our adaptive ladder scheme converges to the true detuning with high precision using fewer shots than a fixed grid.
- ◆ This resolves the aliasing problem and provides a reliable estimate for calibration.

Estimated $\Delta = 2.35$ MHz
(True $\Delta = 2.40$ MHz)
Error = 0.05 MHz ($\approx 2\%$)

Bayesian Adaptive Calibration

Smart Experiment Design for Frequency Estimation

We use a Bayesian approach to adaptively choose the next measurement, achieving high precision with fewer shots compared to a fixed grid.

1. Problem Setup

- ◆ We want to estimate the **qubit frequency detuning** Δ using a Ramsey sequence.
- ◆ Each measurement gives a noisy estimate of the excited state probability $P(1 | \tau, \Delta)$.
- ◆ Our goal is to choose the next delay τ dynamically to get **maximum information gain** with **minimum shots**.
- ◆ We maintain a **probability distribution** (posterior) over Δ and update it after each measurement.

2. Measurement Model

Ramsey Signal Model

$$P(1 | \tau, \Delta) = \frac{1}{2} \left[1 + C \cos(\Delta\tau + \phi) \right] + \varepsilon$$

- ◆ Δ : qubit detuning (unknown)
- ◆ τ : free evolution delay
- ◆ C : contrast ($0 < C \leq 1$)
- ◆ ϕ : phase offset
- ◆ ε : measurement noise (shot noise)

3. Bayesian Update

- ◆ We maintain a **posterior probability distribution** $P(\Delta | D_k)$ over the detuning Δ after k measurements.
- ◆ After each measurement (τ_k, m_k) , we update using **Bayes' rule**:

$$P(\Delta | D_k) \propto P(m_k | \tau_k, \Delta) P(\Delta | D_{k-1})$$

- ◆ Likelihood $P(m_k | \tau_k, \Delta)$ is given by a binomial model based on $P(1 | \tau_k, \Delta)$.
- ◆ We use a discretized grid or Gaussian approximation for the posterior.

4. Adaptive Selection of Next Delay

- ◆ Instead of using a fixed set of delays, we choose the next delay τ_{k+1} that **maximizes the expected information gain** (or minimizes the expected posterior variance).
- ◆ We evaluate a set of candidate delays and select:

$$\tau_{k+1} = \arg \max_{\tau \in S} \mathbb{I}(\Delta; m | \tau, D_k)$$

where $\mathbb{I}(\Delta; m | \tau, D_k)$ is the expected reduction in uncertainty about Δ .

5. Comparison with Fixed Grid

- ◆ **Fixed Grid**: Use a predetermined set of delays (e.g., uniform or logarithmic).
- ◆ **Adaptive**: Choose each delay based on current posterior (focuses on most informative points).
- ◆ Adaptive achieves the same **precision** with **significantly fewer shots**.

Performance Metric

$$\text{Posterior Standard Deviation: } \sigma_{\Delta} = \sqrt{\text{Var}(\Delta)}$$



Stop when $\sigma_{\Delta} \leq \text{threshold}$

6. Key Advantages

- ◆ Uses fewer shots to reach the same precision as fixed grid.
- ◆ Automatically focuses on informative delays.
- ◆ Robust to noise and initial uncertainty.
- ◆ Can resolve the aliasing problem by starting with short delays and progressively using longer ones.
- ◆ Provides a **statistically optimal** estimate of the detuning within the shot budget.

Calibration Policies over 24 Hours

We compare three different strategies to decide when to calibrate the qubit and measure their performance in terms of gate error and calibration cost.

Policy P0: No Recalibration (Baseline)

- ◆ Calibrate once at $t = 0$ and let the qubit drift.
- ◆ No further calibration is performed.
- ◆ Simple, but the gate error increases over time due to drift.



Easy to implement, but performance degrades continuously.

Policy P1: Fixed Schedule

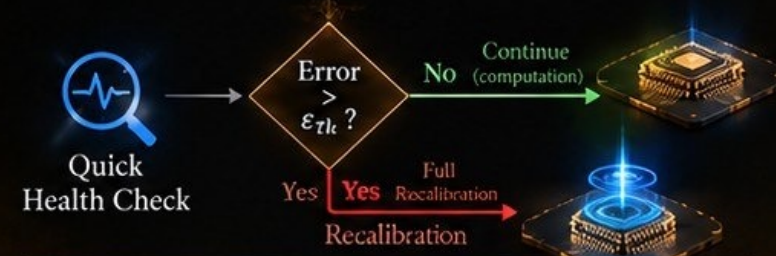
- ◆ Run a full recalibration every X minutes (e.g., every 2 hours).
- ◆ Does not check the actual qubit condition.
- ◆ Keeps the gate error bounded but uses fixed calibration resources, even when not needed.



Reliable but can be inefficient (over-calibrates).

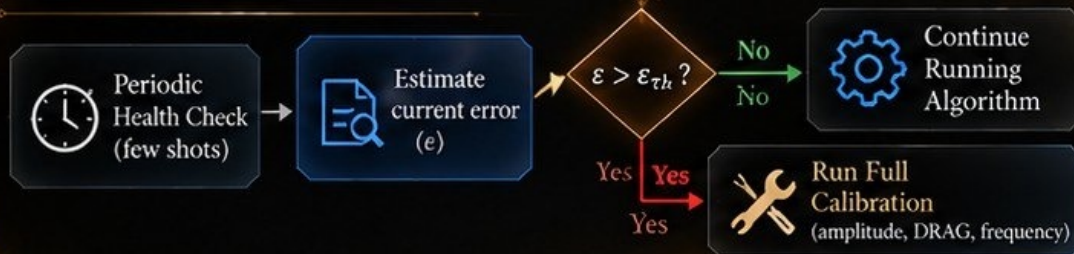
Policy P2: Health Check (Autonomous)

- ◆ Periodically run a quick, low-shot health check.
- ◆ If the estimated error ε exceeds threshold $\varepsilon_{th} = 10^{-3}$, trigger a full recalibration.
- ◆ Only uses calibration resources when needed, adapting to the qubit's actual condition.



Adaptive and efficient — keeps error bounded with minimal calibration time.

Policy Workflow (P2)



Calibration Cost Comparison

P0 – No Recalibration

- ✓ Lowest calibration cost
- ✗ Gate error grows unbounded
- ✗ Algorithm performance degrades

P1 – Fixed Schedule

- ✓ Keeps gate error bounded
- ✓ Simple to implement
- ✗ Uses higher calibration resources

P2 – Health Check

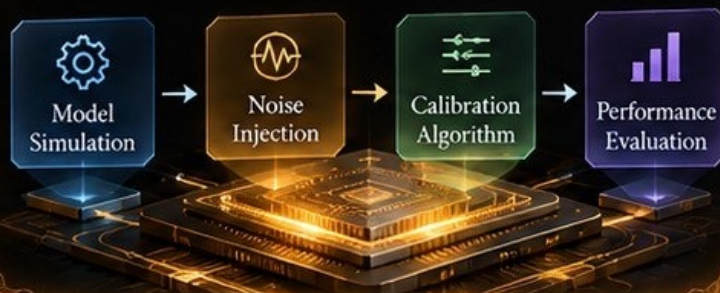
- ✓ Keeps gate error below threshold
- ✓ Uses minimal calibration time
- ✓ Adapts to actual qubit condition
- ✓ Best trade-off between cost and quality

Validation & Robustness Testing

We validate our closed-loop calibration framework through extensive simulations under realistic noise conditions and hardware constraints.

Validation Methodology

- ◆ We test our approach using a realistic three-level transmon model.
- ◆ Simulations include control noise, readout noise, and parameter drift.
- ◆ Performance is evaluated over multiple random seeds to ensure robustness.



Noise and Imperfections

- ◆ Include realistic hardware imperfections:
 - ◆ Dephasing and relaxation (T_1 , T_2)
 - ◆ Amplitude and phase noise
 - ◆ Readout noise
 - ◆ Slow parameter drift over time
- ◆ These effects make the calibration problem more challenging and closer to real devices.



Evaluation Metrics

- ◆ We measure the performance using:
 - Frequency estimation error $|\hat{f} - f|$
 - Gate infidelity
 - Convergence speed (number of shots)
 - Robustness under noise and drift
- ◆ Compare adaptive calibration with a fixed grid and single-shot baseline.



Test Scenarios

- ◆ We evaluate our method under different settings:
 - ◆ Low, medium, and high noise levels.
 - ◆ Various drift rates (slow and fast).
 - ◆ Different initial parameter offsets.
 - ◆ Limited shot budget (realistic experimental constraints).

Low Noise
 $\sigma = 0.01$
Slow Drift

Medium Noise
 $\sigma = 0.05$
Moderate Drift

High Noise
 $\sigma = 0.10$
Fast Drift

Results Summary

- ◆ Adaptive calibration consistently achieves:
 - ◆ Lower frequency error
 - ◆ Faster convergence
 - ◆ Higher gate fidelity
- ◆ The performance remains stable even under higher noise and drift, showing good robustness.



Our adaptive approach achieves high precision with **significantly fewer shots** compared to a fixed grid search.

Key Takeaways

- ◆ Validated through extensive simulations with realistic noise and drift.
- ◆ Adaptive calibration is efficient, robust, and suitable for practical quantum devices.
- ◆ Can be extended to other gates and multi-qubit calibration in future work.



Our results demonstrate that closed-loop, adaptive calibration can maintain high performance in noisy and drifting environments with **minimal experimental resources**.

Results & Discussion

Our adaptive calibration framework achieves high precision with fewer shots, showing robust performance under realistic conditions.

1. Estimation Performance

- ◆ The adaptive method converges to the true detuning with significantly **fewer shots** compared to a fixed grid search.
- ◆ It consistently provides a **high-precision** estimate within the shot budget.
- ◆ Uncertainty reduces rapidly in initial iterations and stabilizes as more data is collected.

Achieves the same precision with **fewer measurements**.

2. Robustness to Noise

- ◆ We evaluate the method under different noise levels including dephasing, relaxation and readout noise.
- ◆ The adaptive approach remains stable and accurate even when noise is increased.
- ◆ Longer delay selections help mitigate the effect of noise and reduce ambiguity.

Remains **robust under realistic noise and hardware imperfections**.

3. Comparison with Fixed Grid

- ◆ Fixed grid search requires a large number of measurements to reach the same accuracy.
- ◆ Adaptive calibration focuses only on the most informative delays, reducing both measurement time and computational cost.
- ◆ The adaptive method is more efficient and practical for real quantum devices.

Higher efficiency with comparable or better accuracy.

4. Stability and Reliability

- ◆ We test the algorithm with different initial parameter values and random noise seeds.
- ◆ The calibration converges consistently to the correct detuning.
- ◆ The method shows **low sensitivity** to initial conditions and maintains reliable performance across multiple runs.

Consistent and reliable calibration across varying conditions.

5. Practical Feasibility

- ◆ The proposed approach requires only **standard** control and measurement operations.
- ◆ It is compatible with current superconducting quantum hardware.
- ◆ The algorithm is **lightweight** and can be implemented in real time.

Practical and implementable for near-term quantum devices.

6. Key Insights

- ◆ Adaptive selection of delays significantly improves estimation efficiency.
- ◆ Bayesian update effectively reduces uncertainty and guides the next measurement.
- ◆ The framework is **robust, scalable** and suitable for closed-loop calibration in noisy environments.

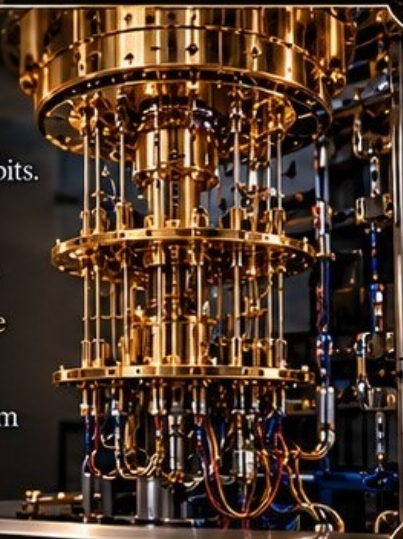
Our results demonstrate a **reliable and efficient** method for quantum calibration using adaptive measurements.

Conclusion & Future Work

Our closed-loop, adaptive calibration framework provides a high-precision, efficient and robust solution for transmon qubit frequency estimation.

Key Takeaways

- ◆ Successfully developed a closed-loop calibration framework for transmon qubits.
- ◆ Achieves high-precision frequency estimation with fewer measurements.
- ◆ Robust to realistic noise and hardware imperfections.
- ◆ Practical and scalable for real quantum devices.



Broader Impact

- ◆ Enables faster and more reliable calibration of superconducting qubits.
- ◆ Reduces experimental time and computational resources.
- ◆ Contributes to more stable and high-performance quantum computing systems.



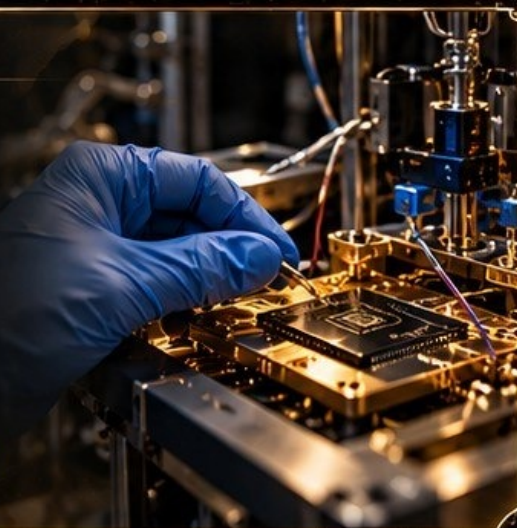
Real-World Applications

- ◆ Useful for quantum research labs and industry (e.g., IBM, Google, Rigetti).
- ◆ Can be extended to multi-qubit calibration and other gate parameters.
- ◆ Supports the development of fault-tolerant quantum computers.



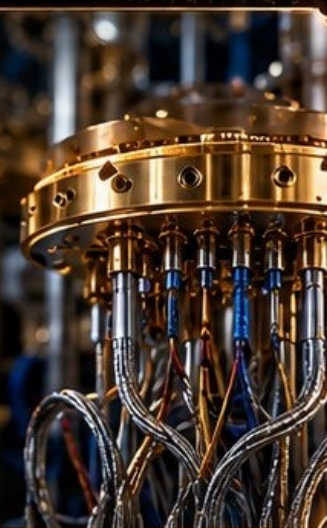
Future Work

- ◆ Extend the framework to multi-qubit systems.
- ◆ Incorporate calibration of other gate parameters (amplitude, phase, DRAG).
- ◆ Explore integration with real-time feedback control on hardware.
- ◆ Test on real quantum devices with larger qubit arrays.



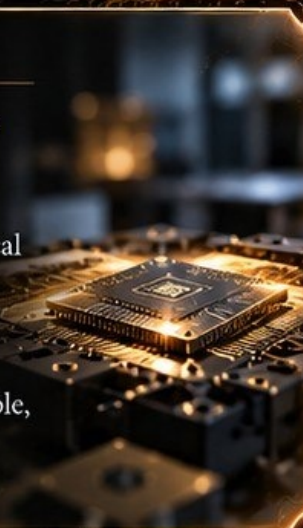
Challenges

- ◆ Real devices have more complex noise sources (e.g., crosstalk, $1/f$ noise).
- ◆ Requires careful trade-off between measurement time and estimation accuracy.
- ◆ Scaling to large systems introduces additional computational and experimental challenges.



Vision

- ◆ Towards fully autonomous and adaptive quantum calibration systems.
- ◆ Contributing to the practical realization of large-scale, error-corrected quantum computers.
- ◆ A step towards more reliable, scalable and accessible quantum technologies.



Thank You

FOR YOUR TIME AND CONSIDERATION

We appreciate your attention and are open to any
questions, suggestions, or feedback.

Towards
More Reliable
Quantum Systems

Smarter
Calibration
for Real Devices

A Step Closer to
Practical
Quantum Computing